

1.21.4. Case Study: Engine Emitting Blue Smoke

Model	Machine No.	Hours (h)	Location	Environment
FR60E	FTC003RLAJX103103	920	Chengdu	Farm, 20-28°C

Symptom: Engine emitting blue smoke.	
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Possible Causes	1. Oil quality issue 2. Insufficient cylinder pressure 3. Fuel injector issue 4. Fuel pump issue
Cause Analysis	Blue smoke, heavy diesel smell, incomplete combustion, or low cylinder pressure, or poor injector atomization.
Troubleshooting Steps	1. Fuel check OK. 2. High-pressure pump OK. 3. Cylinder cutout test: Misfire on cylinder 4. 4. Cylinder pressure tested normal. 5. Diagnosis: Injector issue. Fault fixed by replacement.
Maintenance Plan	Replace fuel injector
Effect Confirmation	Fault disappeared
Prevention / Suggestion	Use high-quality fuel regularly.

1.21.5. Case Study: Severe Idle Vibration

Model	Machine No.	Hours (h)	Location	Environment
FR-60E	FTC003RFVHH102378	1261	Tumxuk, Xinjiang	Earthmoving

Symptom:

Heavy engine shaking when the excavator is idling.



Possible Causes	1. Loose mounting bolts 2. Cylinder misfire 3. Damaged fuel injector 4. Damaged engine rubber mounts
Cause Analysis	1. Loose bolts cause vibration. 2. Misfiring causes power imbalance. 3. Poor injector atomization leads to uneven power. 4. Worn mounts fail to absorb engine tremors.
Troubleshooting Steps	Vibration is severe under 1500 RPM but smooth otherwise. Tightened all bolts (no change). Specialists tested injectors and misfire (no issues). Inspected mounts and found wear. Replaced all 4 rubber mounts. Idle vibration resolved.
Maintenance Plan	Replace engine rubber mounts
Effect Confirmation	Vibration cleared, running normally.
Prevention / Suggestion	Inspect mount condition during scheduled maintenance.

1.21.6. Case Study: High Coolant Temperature Alarm

Model	Machine No.	Hours (h)	Location	Environment
FR60E	FTC003RFTTH10237 3	1519	Macheng	Sand Quarry, 20-28°C

Symptom:
Water temperature alarm triggered at 120°C limit. Occasional short circuit errors.



Possible Causes	Sensor fault, Wiring ground, Poor engine grounding, Dashboard issue, Radiator blockage, Fan belt slip.
Cause Analysis	Differentiate between actual overheating vs. faulty sensor readings.
Troubleshooting Steps	Actual temperature measured at 62°C (False Alarm). Wiring showed no ground but heating the sensor showed correct readings on the dash. Found poor connection at the main power switch which caused erratic electrical signaling. Fault cleared after power switch replacement.
Maintenance Plan	Replace main power master switch
Effect Confirmation	High temperature alarm fixed.
Prevention / Suggestion	Ensure all grounding points are clean and tight.

1.21.7. Case Study: Occasional Hard Starting After Extended Parking

Model	Machine No.	Hours (h)	Location	Environment
FR60E	FTC003RFCLW110055	413	Wuhan	Sand Quarry, 20-28°C

Symptom:

Engine fails to start occasionally after being parked for long periods.



Possible Causes	Stop solenoid, Stop relay, GPS interlock, Low-pressure fuel line blockage, Pump fault.
Cause Analysis	Engine cranks but doesn't fire. No smoke from exhaust during failed start attempts.
Troubleshooting Steps	Replaced fuel filters and separators. Replaced stop solenoid and timer. Problem persisted. Finally cleaned the fuel tank and found rust/water sediment at the bottom causing intermittent fuel flow blockage for the low-pressure pump.
Maintenance Plan	Deep clean the fuel tank
Effect Confirmation	Starting logic restored.
Prevention / Suggestion	Drain water separator daily; keep fuel tank full to avoid condensation.